

Rowan Orlijan-Rhyne

website

Employment

Aug 2025 - **Research assistant, Max Planck Institute for Meteorology (MPI-M)**
supervision: Bjorn Stevens, Tobias Kölling, Lukas Kluff
Production and analysis of continuously growing, analysis-ready datasets from the Barbados Cloud Observatory (BCO) using Python, Apache Airflow.
Poster presentations: May 2026, EGU Annual Meeting, Vienna & Aug 2026, Oeschger Centre for Climate Change Research (OCCR) Swiss Climate Summer School

Education

Aug 2025 - **MPI-M**, seminars and lectures, notably:
Climate Circulations and Energy (lectures from Sarah Kang and guests)
C++ intensive course
Oeschger Centre for Climate Change Research (OCCR) Swiss Climate Summer School in Grindelwald, Switzerland focused on climate variability

Sep 2021 - May 2025 **Swarthmore College**, B.A., Physics, Phi Beta Kappa
Minors: Applied Mathematics and Global Studies
Involvement: NCAA DIII Cross-Country and Track and Field Captain
Attended **Sorbonne Sciences** and **Paris Cité** University (Fall 2023, French language)
Attended **Sapienza University** (Spring 2024, Italian language)

Research Experience

January 2025 - July 2025 **Research Assistant, Department of Physics, Swarthmore College**
supervision: Amy Graves
Can Graph Neural Network (GNN) replace Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) simulations of disordered granular systems?
Virtual conference presentation: March 2026, APS Global Physics Summit

summers 2023 & 2024 **Research Assistant, Climate Modeling Alliance (CliMA), California Institute of Technology**
supervision: Tapio Schneider & Anna Jaruga
Implemented ice-phase microphysics scheme in CloudMicrophysics.jl for Julia-based model, making use of 1-dimensional kinematic framework for testing.
Poster presentation: March 2025, APS Global Physics Summit, Anaheim, California

summer 2022 **NSF REU: Washington State University's Lab for Atmospheric Research (LAR) Air Quality, Atmospheric Chemistry, and Climate Change: Measurements and Modeling in the Pacific Northwest**
supervision: Shelley Pressley & Fabio Scarpare
Used agricultural model CropSyst combined with in-situ observations to study impact of agricultural practices on Brazilian sugar cane production.

Publications

2026 (in revision) **Dynamical regimes reveal drivers of Earth's growing energy imbalance**
C.E. Singer, O. Hadas, M. Schulz, R. Orlijan-Rhyne, F. Glassmeier, F. Bender
pre-print: <https://doi.org/10.22541/essoar.15005264>

2026 **Earth's net incoming energy-flux imbalance (and its implications for understanding climate change)**
Stevens, B., Andrews, T., Bender, F.-A.-M., Bony, S., Ceppi, P., Glassmeier, F., et al.
<https://doi.org/10.17617/2.3711842>

Mentorship/Teaching

Aug 2025 - Spring 2025	Informal supervision of and collaboration with interns and student workers at MPI Electricity, Magnetism, and Optics with Biomedical Applications TA Electricity, Magnetism, and Optics with Biomedical Applications Tutor Linear Algebra Tutor
Fall 2024	Motion, Force, and Energy with Biomedical Applications TA
Spring 2023	Thermodynamics, Statistical Mechanics, and Special Relativity Grader
Fall 2022	Optics and Quantum Mechanics Grader
Spring 2022	Single Variable Calculus II Tutor
Summer 2020	Founder and director of online physics camp for youth

Honors

2025- 2024	Phi Beta Kappa Academic Honors Society Member Virtual Delegate, United Nations Climate Change Conference 29 (COP29)
2021 - 2025	Kiwanis Club of Claremont Stan Larson Scholar Sigma Xi Academic Honors Society Member
2021	International Baccalaureate Diploma, <i>40 pts</i>

Personal

Programming Languages	Python, Julia, C++, Matlab, Arduino, R
Tools	Pandas, Xarray, PyTorch, Tensorflow, SciKit, Apache Airflow
Citizenship	United States
Languages	English, Italian, French, and ein bisschen Deutsch
Interests	running, cooking, reading novels